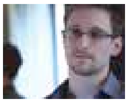


Snowden used Bitcoin servers
Edward Snowden used Bitcoin to buy servers for 2013 mass surveillance leak.
https://dgit.in/july19-94

Second foldable phone?
Samsung is reportedly planning a second foldable phone despite the Fold delay.
https://dgit.in/july19-95

WD Black SN750 NVMe SSD 1 TB

A little bit here and there

The WD SN750 is the third iteration of the WD Black lineup of M.2 SSDs. With the SN750, we should see another high-performer given that they've stuck with the same proprietary controller and even the same 64-layer 3D TLC NAND. We are probably looking at a revision of the controller along with improved firmware in this new SN750 package. The one thing that stands out with the SN750 is the massive EKWB heatsink that it comes with. There are also variants available without the heatsink.

WD didn't share any of the controller specifications. By comparing the spec sheets side-by-side we can see that practically all the specs are the same except for the read and write characteristics. This would indicate that the bulk of the improvements with this SKU lie in the firmware. The rated Seq. read speeds have improved by 70 MBps and the rated Seq. write speeds have gone up by 200 MBps. Random read and write IOPS figures also show incremental improvements. The 2018 WD Black was only available in 250GB/500GB/1TB variants whereas the 2019 SN750 now has a 2TB option as well.

Build and Design

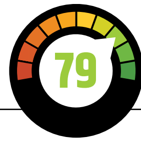
The heatsink is the first thing you notice about the WD Black SN750. The need of a heatsink has often been debated in the analyst circles and the answer lies in how and where the SSD is installed. If there's plenty of incidental air-flow then the threshold for the thermal throttling is rarely attained by most users, if not then there is a very realistic possibility of the SSD throttling. You might want to check the clearance of the WD Black before installing it on your motherboard as it is quite a tall heatsink. The SSD stands 8.10 mm tall with the heatsink installed and there's also the fact that once it is installed, it will sit a little higher on the motherboard as the M.2 port has a little bit of clearance underneath it.

Also, the heatsink wraps around the SSD. So you not only have the added height but also a slight addition to the width. The two put together are actually problematic for certain motherboards. WD is addressing this by putting out a compatibility list on their website highlighting which boards are compatible and which aren't along with which boards would require the heatsink to be removed before it can be installed.

Performance

You might be wondering if there will be a performance difference if the controller as well as NAND chips are kept the same, and the firmware is the key differentiator. You'd be surprised with what all can be achieved with just software/firmware tweaks. When we take a look at CrystalDiskMark, which is a very simple and synthetic benchmark that showcases the raw read and write speeds, you don't see a drastic improvement. CrystalDiskMark can hardly bring out the nuances of firmware improvements but we have it anyway since it happens to be popular. The performance numbers seem close to the spec sheet i.e. around the 3470 MBps mark for read speeds and 2800 MBps write speeds. s

Moving on to analysing the driver performance post conditioning, we start to see the differences between the WD Black 2018 (SN700) and WD SN750 start to emerge.



PERFORMANCE..........87
BUILD 70
VALUE FOR MONEY....79

The WD SN750 scores more than the WD Black SN700 in our sustained 128K read tests by a considerable margin of approximately 360 MBps. This brings it very close to the Intel 900p and the Corsair MP510 which are both excellent performers in their own right. Samsung's 970 EVO is comparable but it should be noted that our 970 EVO was a 512 GB SKU and not a 1 TB SKU as all the other drives that we test are. Even in the sequential write performance test we find the WD750 to show an improvement of about 480 MBps which is quite sizeable as well.

Verdict

With the WD Black SN750 we have an excellent example of how much performance improvement the software along can bring to the table. It's certainly odd that WD did not simply issue these firmware changes for the older SN700 since it too could have easily benefitted from the performance boost and it would have been a very consumer friendly move as well.

–Mithun Mohandas

SPECIFICATIONS

WDS100T3XHC | **FORM FACTOR:** M.2 2280 | **COST/GB:** ₹23.44 | **DIMENSIONS:** 80 x 22 x 8.10 mm | **CONTROLLER:** WD Proprietary, SanDisk 64-layer 3D NAND | **WARRANTY:** 5 years

CONTACT

WD | **PHONE:** 1800-200-5789 | **EMAIL:** Website form | **WEBSITE:** www.wdc.com